SSC MTS 2024 question paper with answers

14 October 2024 · Shift 1

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

A solid cuboid with dimensions 2 m × 4 m × 10 m is melted to form small solid cubes of side 8 cm. How many such cubes can be formed?

(A) 1,60,000
(B) 1,62,250
(C) 1,56,250
(D) 1,58,250

Q.2

Raman paid ₹18,656 for a mobile phone, after a single discount of 12%. If Raman is given successive discounts of 4% and 8%, then how much Raman would pay?

(A) ₹18,723.84
(B) ₹17,283.84
(C) ₹17,238.84
(D) ₹13,728.84

Q.7

A boat can travel at a speed of 28 km/h in still water. If the speed of the stream is 4 km/h, what is the average speed (in km/h) of the boat if it travels 192 km downstream, stops there for one hour and then returns to the starting point?

(A) 25.6
(B) 28.5
(C) 24.8
(D) 36.5

Q.8

Find the value of 5 + 32 ÷ 2 + 6 × 2.5.

(A) 36
(B) 40
(C) 34
(D) 15

Q.9

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 6% per year rate of interest for 5 years.

(A) 790
(B) 750
(C) 690
(D) 710

Q.10

A mixture contains 8 kg of tea costing ₹295 per kilogram and 10 kg of tea costing ₹250 per kilogram. The average price per kilogram (in ₹) of the mixture is:

(A) 270
(B) 273
(C) 259.60
(D) 250.70

Q.11

In a school, the average salary of 12 teachers was ₹71,400. If the average salary of the teachers, principal and vice principal was ₹89,500, then what was the average salary of the principal and vice principal (in ₹)?

(A) 1,76,800
(B) 1,18,900
(C) 1,98,100
(D) 2,11,000

Q.12

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 6% per year rate of interest for 3 years.

(A) 630
(B) 570
(C) 670
(D) 590

Q.15

Find the average of the prime numbers lying between 68 and 91.

(A) 78
(B) 79
(C) 76
(D) 77

Q.16

A rectangular sheet of paper of length 10π cm and width 5 cm is rolled along its length (width of the sheet becomes the height of the cylinder) to form a cylinder. What would be the radius of the cylinder?

(A) 2.5 cm
(B) 5 cm
(C) 6 cm
(D) 10 cm

Q.17

If two numbers are 6.25% and 25% more than a third number, respectively, then the first number is how much per cent of the second number?

(A) 95%
(B) 80%
(C) 85%
(D) 90%

Q.18

A dishonest dealer professes to sell his goods at cost price but uses a false weight and thus gains 15%. For a kilogram he uses a weight of ____ grams.

(A) 830.43
(B) 1130.43
(C) 930.43
(D) 869.57

Q.19

If 2744 identical small spheres are made out of a big sphere of radius 7 cm, then what is the surface area of each small sphere?

(A) π cm^2
(B) 4π cm^2
(C) 3π cm^2
(D) 2π cm^2

Q.20

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 67 kg and 37 kg is:

(A) 50.8
(B) 51.8
(C) 49.8
(D) 48.8

Reasoning Ability and Problem Solving 15 questions

Q.1

What should come in place of the question mark (?) in the given series based on the English alphabetical order? RVQ, OSN, LPK ,IMH, ?

(A) EJF
(B) JEF
(C) FJE
(D) EFJ

Q.2

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: All cocktails are mocktails. Some cocktails are juices. Conclusion (I): Some mocktails are juices. Conclusion (II): Some juices are not cocktails.

(A) Both conclusions I and II follow
(B) Neither conclusion I nor II follow
(C) Only conclusion II follows
(D) Only conclusion I follows

Q.3

EJGL is related to INKP in a certain way based on the English alphabetical order. In the same way, KPMR is related to OTQV. To which of the following is NSPU related, following the same logic?

(A) WRYT
(B) RWYT
(C) WRTY
(D) RWTY

Q.4

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). F is sitting second to the left of B. F is sitting third to the left of D. B is sitting to the immediate right of E. A is sitting third to the left of E. A is sitting to the immediate right of G. What is the position of F with respect to E?

(A) Third to the right
(B) Second to the left
(C) Immediate right
(D) Immediate left

Q.6

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13, can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (18, 3, 54) (21, 4, 84)

(A) (4, 8, 39)
(B) (8, 7, 58)
(C) (14, 2, 28)
(D) (3, 11, 53)

Q.7

In a certain code language, ‘solve the puzzle’ is written as ‘sp di cy’ and ‘puzzle is tricky’ is coded as ‘qp vt di’. How is ‘puzzle’ coded in the given language?

(A) qp
(B) di
(C) cy
(D) sp

Q.8

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All carrots are onions. No onion is a tomato. Conclusions: (I) No carrot is a tomato. (II) All onions are carrots.

(A) Both conclusions I and II follow.
(B) Neither conclusion I nor II follows.
(C) Only conclusion I follows.
(D) Only conclusion II follows.

Q.9

In a certain code language, ' train was late' is coded as 'ch px zt' and 'journey by train' is coded as 'fk zt bu'. How is 'train' coded in the given language?

(A) bu
(B) px
(C) ch
(D) zt

Q.10

What will come in place of the question mark (?) in the following equation if '+' and '×' are interchanged and '–' and '÷' are interchanged? 55 × 16 – 4 ÷7 + 3 = ?

(A) 17
(B) 12
(C) 38
(D) 28

Q.12

In a certain code language, 'GULF' is coded as '7391' and 'FANS' is coded as '2438'.What is the code for 'F' in the given code language?

(A) 4
(B) 1
(C) 3
(D) 2

Q.15

Which two numbers should be interchanged to make the following equation correct? 36÷ 2 + 4× 8 –11× 3 + 28 = 49 (Note: Interchange should be done of the entire number and not individual digits of a given number.)

(A) 2 and 4
(B) 36 and 28
(C) 2 and 3
(D) 4 and 3

Q.17

HLQC is related to KOTF in a certain way based on the English alphabetical order. In the same way, WTAN is related to ZWDQ. To which of the following is FXCM related, following the same logic?

(A) IAEO
(B) IAFP
(C) GZFQ
(D) HZEO

Q.18

P, Q, R, S, T, U and V are sitting around a circular table, facing towards the centre (but not necessarily in the same order). P is an immediate neighbour of both T and U. S is sitting second to the right of P. Immediate neighbours of R are U and Q. S is not an immediate neighbour of U. Who is sitting to the immediate left of Q?

(A) S
(B) U
(C) V
(D) R

Q.19

What should come in place of the question mark (?) in the following series? 97, 88, 85, 76, 73, ?, 61

(A) 72
(B) 70
(C) 65
(D) 64

Q.20

3 is related to 24 following a certain logic. Following the same logic, 4 is related to 60. To which ofthe following is 6 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 216
(B) 236
(C) 210
(D) 230

General Awareness 25 questions

Q.1

Which of the following was authored by Harisenacharya?

(A) Brihatkatha Kosh
(B) Lilavati
(C) Rajatarangini
(D) Brihat Katha Manjari

Q.2

The state religion of Bangladesh is:

(A) Islam
(B) Buddhism
(C) Christianity
(D) Hinduism

Q.3

What is the major purpose of the Salal Dam located on Chenab River in Jammu and Kashmir?

(A) Drinking
(B) Navigation
(C) Hydroelectricity
(D) Irrigation

Q.4

Which of the following phyla contains Leech?

(A) Coelenterata
(B) Platyhelminthes
(C) Nematoda
(D) Annelida

Q.5

Who among the following was ‘Senani’ in the Rig Vedic period?

(A) A tax officer
(B) A war hero
(C) A spy
(D) A priest

Q.6

In July 2023, Tribhuvaneshwar Saran Singh Deo was appointed as Deputy Chief Minister of which of the following states?

(A) Jharkhand
(B) Chhattisgarh
(C) West Bengal
(D) Odisha

Q.7

In weight lifting, when an athlete misses all three attempts in the snatch or the clean and jerk, failing to post a total, it is known as _____.

(A) Bombing-out
(B) Take-off
(C) Attempt
(D) Delivery

Q.8

The Moplah Uprising, an organised opposition of Muslim cultivators, occurred in which modern state of India?

(A) Kerala
(B) Bihar
(C) Punjab
(D) Tamil Nadu

Q.9

The union list from the Seventh Schedule of the Indian Constitution consists of how many subjects as of May 2024?

(A) Hundred
(B) Ninety five
(C) Ninety three
(D) Ninety nine

Q.10

In February 2023, there have been protests in Punjab to release a group of Sikh prisoners who have been in jail for more than 30 years since militancy was wiped out in Punjab in the 1990s. What are these prisoners called?

(A) Bandi Singhs
(B) Babbar Bandis
(C) Khalsa Tigers
(D) Qaidi Sikhs

Q.11

Which among the following is considered the first experiment of Satyagraha by Gandhiji in India?

(A) Ahmedabad
(B) Champaran
(C) Kheda
(D) Rowlatt Satyagraha

Q.12

Bhavnath Mahadev fair is held every year in which of the following places?

(A) Jabalpur
(B) Chennai
(C) Agra
(D) Junagarh

Q.13

As per the 2001 Census, what is the specific age, for a person to be called literate?

(A) 6 years
(B) 7 years
(C) 9 years
(D) 8 years

Q.14

A section of an active fault that has produced earthquakes in the past but is now quiet and is believed that the fault segment is capable of producing earthquake on some other basis is known as:

(A) Seismic Gap
(B) Epicenter
(C) Seismic Wave
(D) Hypocenter

Q.15

When was the Wildlife Protection Act passed?

(A) 1980
(B) 1951
(C) 1955
(D) 1972

Q.16

Which of the following is a traditional game of India?

(A) Basketball
(B) Football
(C) Kabaddi
(D) Volleyball

Q.17

Who became the first folk artist from India to walk the Cannes red carpet in 2022?

(A) Sarwar Khan
(B) Swaroop Khan
(C) Mame Khan
(D) Sartaj Khan

Q.18

Which of the following statements about Viharas is/are correct? a)Viharas consist of a verandas, cell and hall around the walls of the hall. b)Ajanta cave and Nashik cave are important parts of Vihara architecture.

(A) Neither a nor b
(B) Only a
(C) Only b
(D) Both a and b

Q.19

Who amongst the following stated the Grameen model of microfinance?

(A) Prof. Abhijit Banerjee
(B) Prof. Muhammad Yunus
(C) Dr. Manmohan Singh
(D) Prof. Amartya Sen

Q.20

Anita Sharma, Sangeet Natak Akademi Award winner, is an exponent of _____ dance form.

(A) Sattriya
(B) Kathakali
(C) Mohiniyattam
(D) Odissi

Q.21

Who is the author of the book called 'Beneath the Surface: My Story'?

(A) Milkha Singh
(B) Michael Phelps
(C) Tiger Woods
(D) Kobe Bright

Q.22

Sonal Mansingh is a famous proponent of which of the following dance forms?

(A) Odissi
(B) Manipuri
(C) Kathak
(D) Mohiniyattam

Q.23

The Mahananda River rising in the _____ is a tributary of Ganga.

(A) Kaimur hills
(B) Darjeeling hills
(C) Nanda Devi peak
(D) Chota Nagpur plateau

Q.24

A group of natural substances with variable phenolic structures, which are found in fruits, vegetables, and grains is called:

(A) Flavonoids
(B) Pectins
(C) Hopanoids
(D) Resins

Q.25

If four resistances of 4 ohm each are connected in parallel, what will be their effective resistance?

(A) 16 ohm
(B) 1 ohm
(C) 10 ohm
(D) 4 ohm

English Language and Comprehension 23 questions

Q.1

Select the most appropriate option to fill in the blank. Leisure activities are activities that people _____ in during their free time while they are not working.

(A) evaluate
(B) engage
(C) encourage
(D) exempt

Q.2

Select the INCORRECTLY spelt word.

(A) Fantastic
(B) Forgetfull
(C) Damaged
(D) Expensive

Q.3

Select the correct spelling of the underlined word in the given sentence. The weather was "absolutley" perfect for the outdoor event.

(A) absoultely
(B) absolutly
(C) absolutely
(D) absoluttley

Q.4

Select the most appropriate synonym of the given word. Corroborate

(A) Support
(B) Contend
(C) Elaborate
(D) Suggest

Q.5

Select the option that has used the correct tense.

(A) By the time we arriving at the party, they had been waiting for over an hour.
(B) By the time we arrive at the party, they had been waiting for over an hour.
(C) By the time we arrived at the party, they had been waiting for over an hour.
(D) By the time we arrives at the party, they had been waiting for over an hour.

Q.6

Parts of the following sentence have been given as options. Select the option that contains an error. I have never seen so beautiful a landscape than this one.

(A) so beautiful
(B) I have never seen
(C) a landscape
(D) than this one

Q.8

Select the most appropriate ANTONYM of the given word. Searing

(A) Thrilling
(B) Mocking
(C) Pressing
(D) Freezing

Q.9

Select the most appropriate ANTONYM of the given word. Meticulous

(A) Persevering
(B) Regardless
(C) Exemplary
(D) Determined

Q.10

Recognise the sentence with the correct transitive verb form of break. A. The cyclone broke the window. B.The window broke. C. The cyclone caused the breaking of window. D.The cyclone caused the broken window.

(A) C
(B) D
(C) B
(D) A

Q.11

Select the most appropriate option to fill in the blank. Prem is an excellent chess player. He is the _____ student of the class too.

(A) most intelligent
(B) intelligently
(C) intelligent
(D) more intelligent

Q.12

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. It is a common perception / that the rich sometimes / despise poor.

(A) that the rich sometimes
(B) No error
(C) It is a common perception
(D) despise poor.

Q.13

Select the most appropriate idiom that can substitute the underlined words in the given sentence. His sudden resignation left the company “in a very difficult situation and to have to make a hard decision” as they scrambled to find a replacement.

(A) between a rock and a hard place
(B) with a silver lining
(C) in the dark
(D) on cloud nine

Q.15

Select the most appropriate meaning of the given idiom. an open book

(A) someone who gives advice based on theory not practice
(B) a person of whom her mother is extremely proud
(C) a helpful person
(D) a person or thing that is easy to learn about and understand

Q.16

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Sushil, who recently joined our school / was awarded the Gold Cup / as he worked hardly / throughout the year.

(A) Sushil, who recently joined our school
(B) throughout the year.
(C) as he worked hardly
(D) was awarded the Gold Cup

Q.17

Select the most appropriate ANTONYM of the given word. Boisterous

(A) Buoyant
(B) Noisy
(C) Benevolent
(D) Quiet

Q.18

Select the option that can be used as a one-word substitute for the given group of words. A post without any remuneration

(A) Honorary
(B) Effeminate
(C) Illegible
(D) Bigot

Q.19

Identify the INCORRECTLY spelt word in the given sentence. BBC Media has shown exponenteal growth in their revenue this year.

(A) media
(B) year
(C) revenue
(D) exponenteal

Q.20

Select the option that can be used as a one-word substitute for the given phrase. Belonging or pertaining to an individual from birth

(A) Congenital
(B) Congenial
(C) Congruent
(D) Congregational

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Though people are becoming health-conscious these days, (1)_____ simple lifestyle habits they are prone to fatal diseases. (2)_____ a new study published in 'The Journal of the American College of Cardiology', (3)_____ food more sparingly is linked to a lower risk of heart disease, heart failure , and ischemic heart disease. A significant risk factor for cardiovascular disease, high blood pressure, is demonstrated to be correlated (4)_____ high sodium intake. As per the research, a person's individual sodium intake over time may be predicted by how (5)____ they salt their food. It is found that people who do not put a little additional salt to their food very often had a much lower risk of heart disease event regardless of lifestyle factors and pre-existing disease.

Select the most appropriate option to fill in blank number 1.

(A) because
(B) for
(C) because of
(D) but

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Though people are becoming health-conscious these days, (1)_____ simple lifestyle habits they are prone to fatal diseases. (2)_____ a new study published in 'The Journal of the American College of Cardiology', (3)_____ food more sparingly is linked to a lower risk of heart disease, heart failure , and ischemic heart disease. A significant risk factor for cardiovascular disease, high blood pressure, is demonstrated to be correlated (4)_____ high sodium intake. As per the research, a person's individual sodium intake over time may be predicted by how (5)____ they salt their food. It is found that people who do not put a little additional salt to their food very often had a much lower risk of heart disease event regardless of lifestyle factors and pre-existing disease.

Select the most appropriate option to fill in blank number 2.

(A) According to
(B) According with
(C) In according with
(D) Accordingly

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Though people are becoming health-conscious these days, (1)_____ simple lifestyle habits they are prone to fatal diseases. (2)_____ a new study published in 'The Journal of the American College of Cardiology', (3)_____ food more sparingly is linked to a lower risk of heart disease, heart failure , and ischemic heart disease. A significant risk factor for cardiovascular disease, high blood pressure, is demonstrated to be correlated (4)_____ high sodium intake. As per the research, a person's individual sodium intake over time may be predicted by how (5)____ they salt their food. It is found that people who do not put a little additional salt to their food very often had a much lower risk of heart disease event regardless of lifestyle factors and pre-existing disease.

Select the most appropriate option to fill in blank number 3.

(A) salt with
(B) salt
(C) salted
(D) salting

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Though people are becoming health-conscious these days, (1)_____ simple lifestyle habits they are prone to fatal diseases. (2)_____ a new study published in 'The Journal of the American College of Cardiology', (3)_____ food more sparingly is linked to a lower risk of heart disease, heart failure , and ischemic heart disease. A significant risk factor for cardiovascular disease, high blood pressure, is demonstrated to be correlated (4)_____ high sodium intake. As per the research, a person's individual sodium intake over time may be predicted by how (5)____ they salt their food. It is found that people who do not put a little additional salt to their food very often had a much lower risk of heart disease event regardless of lifestyle factors and pre-existing disease.

Select the most appropriate option to fill in blank number 4.

(A) of
(B) for
(C) in
(D) with

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Though people are becoming health-conscious these days, (1)_____ simple lifestyle habits they are prone to fatal diseases. (2)_____ a new study published in 'The Journal of the American College of Cardiology', (3)_____ food more sparingly is linked to a lower risk of heart disease, heart failure , and ischemic heart disease. A significant risk factor for cardiovascular disease, high blood pressure, is demonstrated to be correlated (4)_____ high sodium intake. As per the research, a person's individual sodium intake over time may be predicted by how (5)____ they salt their food. It is found that people who do not put a little additional salt to their food very often had a much lower risk of heart disease event regardless of lifestyle factors and pre-existing disease.

Select the most appropriate option to fill in blank number 5.

(A) soundly
(B) hefty
(C) frequently
(D) burly

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.2 **(A)**	Q.7 **(A)**	Q.8 **(A)**	Q.9 **(B)**	Q.10 **(A)**
Q.11 **(C)**	Q.12 **(A)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(C)**	Q.18 **(D)**
Q.19 **(A)**	Q.20 **(C)**				

Reasoning Ability and Problem Solving

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(D)**	Q.6 **(C)**	Q.7 **(B)**
Q.8 **(C)**	Q.9 **(D)**	Q.10 **(C)**	Q.12 **(C)**	Q.15 **(B)**	Q.17 **(B)**
Q.18 **(C)**	Q.19 **(D)**	Q.20 **(C)**			

General Awareness

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(C)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(B)**
Q.7 **(A)**	Q.8 **(A)**	Q.9 **(A)**	Q.10 **(A)**	Q.11 **(B)**	Q.12 **(D)**
Q.13 **(B)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(C)**	Q.17 **(C)**	Q.18 **(D)**
Q.19 **(B)**	Q.20 **(A)**	Q.21 **(B)**	Q.22 **(A)**	Q.23 **(B)**	Q.24 **(A)**
Q.25 **(B)**					

English Language and Comprehension

Q.1 **(B)**	Q.2 **(B)**	Q.3 **(C)**	Q.4 **(A)**	Q.5 **(C)**	Q.6 **(D)**
Q.8 **(D)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(A)**	Q.12 **(D)**	Q.13 **(A)**
Q.15 **(D)**	Q.16 **(C)**	Q.17 **(D)**	Q.18 **(A)**	Q.19 **(D)**	Q.20 **(A)**
Q.21 **(C)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(D)**	Q.25 **(C)**	

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) 1,56,250

The cuboid is 200 × 400 × 1,000 = 8,00,00,000 cm^3 and each cube 8^3 = 512 cm^3. 8,00,00,000 ÷ 512 = 1,56,250.

Q.2 · Answer (A) ₹18,723.84

The marked price is 18,656 ÷ 0.88 = ₹21,200. After 4% and then 8% off: 21,200 × 0.96 × 0.92 = ₹18,723.84.

Q.7 · Answer (A) 25.6

Downstream at 28 + 4 = 32 km/h takes 192/32 = 6 hours; upstream at 28 − 4 = 24 km/h takes 8 hours. With the hour's stop, 384 km takes 15 hours: 384/15 = 25.6 km/h.

Q.8 · Answer (A) 36

Division and multiplication first: 5 + 16 + 15 = 36.

Q.9 · Answer (B) 750

SI = 2500 × 6 × 5/100 = ₹750.

Q.10 · Answer (A) 270

(8 × 295 + 10 × 250)/18 = (2,360 + 2,500)/18 = 4,860/18 = ₹270.

Q.11 · Answer (C) 1,98,100

All 14 together earned 14 × 89,500 = ₹12,53,000 and the 12 teachers 12 × 71,400 = ₹8,56,800. The principal and vice principal earned the other ₹3,96,200: an average of ₹1,98,100.

Q.12 · Answer (A) 630

SI = 3500 × 6 × 3/100 = ₹630.

Q.15 · Answer (B) 79

The primes between 68 and 91 are 71, 73, 79, 83 and 89. Their sum is 395, so the average is 395/5 = 79.

Q.16 · Answer (B) 5 cm

Rolled along its length, the 10π cm side becomes the circumference: 2πr = 10π, so r = 5 cm.

Q.17 · Answer (C) 85%

Taking the third number as 100, the two numbers are 106.25 and 125. 106.25/125 × 100 = 85%.

Q.18 · Answer (D) 869.57

Charging for 1,000 g while giving w g gains (1,000 − w)/w = 15%, so w = 1,000/1.15 = 869.57 grams.

Q.19 · Answer (A) π cm^2

Volume goes with the cube of the radius: 2744 = 14^3, so each small sphere has radius 7/14 = 0.5 cm. Its surface area is 4π × 0.5^2 = π cm^2.

Q.20 · Answer (C) 49.8

(40 + 49 + 56 + 67 + 37)/5 = 249/5 = 49.8 kg.

Reasoning Ability and Problem Solving

Q.1 · Answer (C) FJE

Every letter moves 3 places back each time: R, O, L, I, F; V, S, P, M, J; Q, N, K, H, E. The next cluster is FJE.

Q.2 · Answer (D) Only conclusion I follows

The cocktails that are juices are also mocktails, so some mocktails are juices (I). Every juice may be a cocktail, so II does not follow.

Q.3 · Answer (D) RWTY

Each letter moves 4 places forward: E → I, J → N, G → K, L → P and K → O, P → T, M → Q, R → V. So NSPU → RWTY.

Q.4 · Answer (D) Immediate left

Facing the centre, a person's left is the next seat clockwise. Going clockwise from E the order is E, F, C, A, G, D, B: B is to E's immediate right, F second to the left of B and third to the left of D, A third to the left of E and to G's immediate right. So F is to E's immediate left.

Q.6 · Answer (C) (14, 2, 28)

In each set the third number is the product of the first two: 18 × 3 = 54 and 21 × 4 = 84. Likewise 14 × 2 = 28.

Q.7 · Answer (B) di

'puzzle' is the only word in both sentences and 'di' the only code in both, so puzzle = di.

Q.8 · Answer (C) Only conclusion I follows.

All carrots are onions and no onion is a tomato, so no carrot is a tomato (I). 'All carrots are onions' does not make all onions carrots, so II does not follow.

Q.9 · Answer (D) zt

'train' is the only word in both sentences and 'zt' the only code in both, so train = zt.

Q.10 · Answer (C) 38

After interchanging the signs: 55 + 16 ÷ 4 – 7 × 3 = 55 + 4 – 21 = 38.

Q.12 · Answer (C) 3

F is the only letter in both words and 3 the only digit in both codes, so F = 3.

Q.15 · Answer (B) 36 and 28

Interchanging 36 and 28: 28 ÷ 2 + 4 × 8 – 11 × 3 + 36 = 14 + 32 – 33 + 36 = 49.

Q.17 · Answer (B) IAFP

Each letter moves 3 places forward: H → K, L → O, Q → T, C → F and W → Z, T → W, A → D, N → Q. So FXCM → IAFP (X wraps round to A).

Q.18 · Answer (C) V

Facing the centre, a person's left is the next seat clockwise. S is second to P's right, so P's right-hand neighbour is T (U may not sit next to S) and U is on P's left. Clockwise from P: P, U, R, Q, V, S, T, with R between U and Q. V is to Q's immediate left.

Q.19 · Answer (D) 64

The differences alternate –9 and –3: 97, 88, 85, 76, 73, 64, 61.

Q.20 · Answer (C) 210

Each number goes to its cube minus itself: $3^3 – 3 = 24$ and $4^3 – 4 = 60$. So $6^3 – 6 = 210$.

General Awareness

Q.1 · Answer (A) Brihatkatha Kosh

The Brihatkatha Kosha, a collection of Jain stories, was written by the Jain monk Harisena (Harisenacharya) in the 10th century. Lilavati is Bhaskara II's, the Rajatarangini Kalhana's and the Brihat Katha Manjari Kshemendra's.

Q.2 · Answer (A) Islam

Bangladesh's Constitution makes Islam the state religion (Article 2A), while guaranteeing equal status to the other religions.

Q.3 · Answer (C) Hydroelectricity

The Salal project on the Chenab, in Reasi district, is a run-of-the-river hydroelectric project of NHPC.

Q.4 · Answer (D) Annelida

The leech is a segmented worm, like the earthworm, and belongs to the phylum Annelida.

Q.5 · Answer (B) A war hero

In the Rig Vedic tribe the senani led the army in war; the priest was the purohita.

Q.6 · Answer (B) Chhattisgarh
T. S. Singh Deo was made Chhattisgarh's first Deputy Chief Minister in mid-2023, under Chief Minister Bhupesh Baghel.

Q.7 · Answer (A) Bombing-out
A weightlifter who fails all three attempts in either lift posts no total and is said to have 'bombed out'.

Q.8 · Answer (A) Kerala
The Moplah (Mappila) uprising of 1921 took place in Malabar, now part of Kerala.

Q.9 · Answer (A) Hundred
SSC's key gives 100, the figure many study guides quote for the Union List (97 subjects originally). Counts differ by source, because entries have since been added and removed: the GST amendment of 2016, for one, dropped entries 92 and 92C.

Q.10 · Answer (A) Bandi Singhs
'Bandi Singhs' is the name given to Sikh prisoners held since the years of militancy in Punjab, whose release Sikh groups have long demanded.

Q.11 · Answer (B) Champaran
Gandhiji's first satyagraha in India was at Champaran in Bihar in 1917, for the indigo farmers; Ahmedabad and Kheda followed in 1918.

Q.12 · Answer (D) Junagarh
The Bhavnath Mahadev fair is held at Mahashivratri at the Bhavnath temple at the foot of Mount Girnar, Junagadh (Gujarat).

Q.13 · Answer (B) 7 years
In the 2001 Census a person aged 7 or above who can read and write with understanding in any language counts as literate.

Q.14 · Answer (A) Seismic Gap
A seismic gap is a stretch of an active fault that has slipped in past earthquakes but has been quiet for a long time, and is expected to produce one again.

Q.15 · Answer (D) 1972
The Wildlife (Protection) Act was passed in 1972.

Q.16 · Answer (C) Kabaddi
Kabaddi is an ancient Indian game; basketball, football and volleyball came from abroad.

Q.17 · Answer (C) Mame Khan
Mame Khan, a Rajasthani folk singer, walked the red carpet at the 2022 Cannes Film Festival as part of India's official delegation.

Q.18 · Answer (D) Both a and b
Viharas, the monks' dwellings, have a hall with cells cut into the walls around it and a veranda in front; the caves at Ajanta and Nashik include well-known viharas.

Q.19 · Answer (B) Prof. Muhammad Yunus
Muhammad Yunus founded the Grameen Bank in Bangladesh, which lends small sums to the rural poor without collateral; he and the bank shared the 2006 Nobel Peace Prize.

Q.20 · Answer (A) Sattriya
Anita Sharma, a Sattriya dancer from Assam, received the Sangeet Natak Akademi Award for Sattriya for 2014.

Q.21 · Answer (B) Michael Phelps
'Beneath the Surface: My Story' is the autobiography of the American swimmer Michael Phelps.

Q.22 · Answer (A) Odissi
Sonal Mansingh is best known for Odissi, though she also trained in Bharatanatyam.

Q.23 · Answer (B) Darjeeling hills
The Mahananda rises in the Darjeeling hills and flows through West Bengal and Bihar to join the Ganga from the left, at Godagari just inside Bangladesh.

Q.24 · Answer (A) Flavonoids
Flavonoids are plant compounds with variable phenolic structures, found in fruits, vegetables, grains, tea and wine.

Q.25 · Answer (B) 1 ohm
In parallel, 1/R = 1/4 + 1/4 + 1/4 + 1/4 = 1, so R = 1 ohm.

English Language and Comprehension

Q.1 · Answer (B) engage
People 'engage in' an activity: they take part in it.

Q.2 · Answer (B) Forgetfull
The correct spelling is 'forgetful', with one l.

Q.3 · Answer (C) absolutely
The correct spelling is 'absolutely'.

Q.4 · Answer (A) Support
To 'corroborate' a statement is to confirm or 'support' it.

Q.5 · Answer (C) By the time we arrived at the party, they had been waiting for over an hour.
The past perfect continuous 'had been waiting' goes with the simple past in the time clause: 'By the time we arrived'.

Q.6 · Answer (D) than this one
'So … a' is followed by 'as', not 'than': 'so beautiful a landscape as this one'.

Q.8 · Answer (D) Freezing
'Searing' means extremely hot; its opposite is 'freezing'.

Q.9 · Answer (B) Regardless
'Meticulous' means very careful about detail. 'Regardless', in its sense of heedless or careless, is its opposite; the other three are terms of praise.

Q.10 · Answer (D) A
A transitive verb takes an object: in 'The cyclone broke the window', 'broke' has the object 'the window'. In B 'broke' has no object, and C and D use a noun or a participle, not the verb.

Q.11 · Answer (A) most intelligent
'The' before the blank and the comparison with the whole class call for the superlative: 'the most intelligent student'.

Q.12 · Answer (D) despise poor.
'The' with an adjective names a whole class of people: 'despise the poor', like 'the rich' earlier in the sentence.

Q.13 · Answer (A) between a rock and a hard place
To be 'between a rock and a hard place' is to face a choice between two difficult options.

Q.15 · Answer (D) a person or thing that is easy to learn about and understand
Someone who is 'an open book' hides nothing and is easy to understand.

Q.16 · Answer (C) as he worked hardly
'Hardly' means scarcely; the adverb for working with effort is 'hard': 'as he worked hard'.

Q.17 · Answer (D) Quiet
'Boisterous' means noisy and lively; its opposite is 'quiet'.

Q.18 · Answer (A) Honorary
An honorary post is one held without pay.

Q.19 · Answer (D) exponenteal
The correct spelling is 'exponential'.

Q.20 · Answer (A) Congenital
A congenital condition is present from birth; 'congenial' means pleasant.

Q.21 · Answer (C) because of
A noun phrase ('simple lifestyle habits') needs the preposition 'because of'; 'because' introduces a clause.

Q.22 · Answer (A) According to
'According to a new study' names the source of the finding.

Q.23 · Answer (D) salting
The gerund 'salting' is the subject: 'salting food more sparingly is linked to a lower risk'.

Q.24 · Answer (D) with
The verb pairs with 'with': 'correlated with high sodium intake'.

Q.25 · Answer (C) frequently
How 'frequently' they salt their food: how often, as the passage says next ('very often').

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